

Calculus III Four Practice Tests - Worked Solutions

Practice Test 1

Question 1

- a) The function is continuous at $(1, -2)$ since the denominator is nonzero:

$$\lim_{(x,y) \rightarrow (1,-2)} \frac{x^2y + y^2}{x^2 + y^2} = \frac{1^2(-2) + (-2)^2}{1^2 + (-2)^2} = \frac{2}{5}.$$

- b) Along $y = mx^2$,

$$\frac{x^2y}{x^4 + y^2} = \frac{mx^4}{x^4 + m^2x^4} = \frac{m}{1 + m^2}.$$

This depends on m , so the limit does not exist.

- c) For $(x, y) \neq (0, 0)$, write $x = r \cos \theta$, $y = r \sin \theta$:

$$\frac{x^2y^2}{x^2 + y^2} = r^2 \cos^2 \theta \sin^2 \theta \rightarrow 0.$$

Thus the function is continuous at the origin when

$$\boxed{a = 0}.$$

Question 2

- a)

$$z = x^2 \cos y + ye^x, \quad x = s^2 + t, \quad y = st.$$

At $(s, t) = (1, 0)$, $(x, y) = (1, 0)$. Also

$$z_x = 2x \cos y + ye^x, \quad z_y = -x^2 \sin y + e^x.$$

So at $(1, 0)$,

$$z_x = 2, \quad z_y = e.$$

Also

$$x_s = 2s = 2, \quad y_s = t = 0, \quad x_t = 1, \quad y_t = s = 1.$$

Therefore

$$\boxed{\frac{\partial z}{\partial s} = z_x x_s + z_y y_s = 2(2) + e(0) = 4}$$

and

$$\boxed{\frac{\partial z}{\partial t} = z_x x_t + z_y y_t = 2(1) + e(1) = 2 + e}.$$

b) Let

$$F(x, y, z) = x^2 + y^2 + z^2 + xyz - 7 = 0.$$

Then

$$F_x = 2x + yz, \quad F_y = 2y + xz, \quad F_z = 2z + xy.$$

By implicit differentiation,

$$\boxed{\frac{\partial z}{\partial x} = -\frac{2x + yz}{2z + xy}}, \quad \boxed{\frac{\partial z}{\partial y} = -\frac{2y + xz}{2z + xy}}.$$

c)

$$f(x, y) = x^2 e^y, \quad \nabla f = \langle 2xe^y, x^2 e^y \rangle.$$

At $(1, 0)$,

$$\nabla f(1, 0) = \langle 2, 1 \rangle.$$

The direction from $(1, 0)$ to $(4, 4)$ is

$$\langle 3, 4 \rangle, \quad \mathbf{u} = \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle.$$

Thus

$$D_{\mathbf{u}}f(1, 0) = \langle 2, 1 \rangle \cdot \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle = \boxed{2}.$$

Question 3

$$f(x, y) = x^2 + y^2 - 2x + 4y = (x - 1)^2 + (y + 2)^2 - 5.$$

The center of the level circles is $(1, -2)$, which lies inside the disk $x^2 + y^2 \leq 16$ because $1^2 + (-2)^2 = 5 < 16$. Hence the minimum occurs at $(1, -2)$:

$$\boxed{f_{\min} = -5}.$$

The maximum occurs on the boundary in the direction farthest from $(1, -2)$, namely

$$(x, y) = -4 \frac{(1, -2)}{\sqrt{5}} = \left(-\frac{4}{\sqrt{5}}, \frac{8}{\sqrt{5}} \right).$$

The farthest distance from $(1, -2)$ to the disk is $4 + \sqrt{5}$, so

$$f_{\max} = (4 + \sqrt{5})^2 - 5 = \boxed{16 + 8\sqrt{5}}.$$

Question 4

The region is enclosed by $x = y^2$ and $x = 2y$. The intersections satisfy

$$y^2 = 2y \implies y = 0, 2.$$

For $0 \leq y \leq 2$,

$$y^2 \leq x \leq 2y.$$

Thus

$$V = \int_0^2 \int_{y^2}^{2y} (2 + x + y^2) dx dy.$$

Integrating,

$$V = \int_0^2 \left[(2 + y^2)x + \frac{x^2}{2} \right]_{y^2}^{2y} dy = \int_0^2 \left(4y + 2y^3 - \frac{3}{2}y^4 \right) dy.$$

Therefore

$$V = \left[2y^2 + \frac{1}{2}y^4 - \frac{3}{10}y^5 \right]_0^2 = 16 - \frac{48}{5} = \boxed{\frac{32}{5}}.$$

Practice Test 2

Question 1

a)

$$\lim_{(x,y) \rightarrow (2,1)} \frac{xy^2 - 2}{x + y - 2} = \frac{2(1)^2 - 2}{2 + 1 - 2} = \boxed{0}.$$

b) Use polar coordinates:

$$\frac{x^3 - 3xy^2}{x^2 + y^2} = \frac{r^3(\cos^3 \theta - 3 \cos \theta \sin^2 \theta)}{r^2} = r(\cos^3 \theta - 3 \cos \theta \sin^2 \theta) \rightarrow 0.$$

So the limit exists and equals

$$\boxed{0}.$$

c)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(x^2 + y^2)}{x^2 + y^2} = 1.$$

Thus

$$\boxed{a = 1}.$$

Question 2

a) Since

$$x = u \cos v, \quad y = u \sin v,$$

we have

$$x^2 + y^2 = u^2.$$

Therefore

$$w = \ln(u^2) = 2 \ln u.$$

Thus

$$\boxed{\frac{\partial w}{\partial u} = \frac{2}{u} = 1 \text{ at } u = 2}, \quad \boxed{\frac{\partial w}{\partial v} = 0}.$$

b) Let

$$F(x, y, z) = e^{xz} + y^2z - x - 1 = 0.$$

Then

$$F_x = ze^{xz} - 1, \quad F_y = 2yz, \quad F_z = xe^{xz} + y^2.$$

Thus

$$\boxed{\frac{\partial z}{\partial x} = \frac{1 - ze^{xz}}{xe^{xz} + y^2}}, \quad \boxed{\frac{\partial z}{\partial y} = -\frac{2yz}{xe^{xz} + y^2}}.$$

c)

$$f(x, y) = \ln(x^2 + y^2), \quad \nabla f = \left\langle \frac{2x}{x^2 + y^2}, \frac{2y}{x^2 + y^2} \right\rangle.$$

At $(1, 1)$,

$$\nabla f(1, 1) = \langle 1, 1 \rangle.$$

The unit vector in the direction $\mathbf{v} = (-3, 4)$ is

$$\mathbf{u} = \left\langle -\frac{3}{5}, \frac{4}{5} \right\rangle.$$

Hence

$$D_{\mathbf{u}}f(1, 1) = \langle 1, 1 \rangle \cdot \left\langle -\frac{3}{5}, \frac{4}{5} \right\rangle = \boxed{\frac{1}{5}}.$$

Question 3

On the boundary $x^2 + y^2 = 25$, use Lagrange multipliers:

$$\nabla f = \lambda \nabla g, \quad \langle 3, -4 \rangle = \lambda \langle 2x, 2y \rangle.$$

Thus

$$3 = 2\lambda x, \quad -4 = 2\lambda y.$$

So

$$x = \frac{3}{2\lambda}, \quad y = -\frac{4}{2\lambda}.$$

Substitute into $x^2 + y^2 = 25$:

$$\frac{9}{4\lambda^2} + \frac{16}{4\lambda^2} = 25 \implies \lambda = \pm \frac{1}{2}.$$

If $\lambda = 1/2$, then $(x, y) = (3, -4)$ and

$$f(3, -4) = 25.$$

If $\lambda = -1/2$, then $(x, y) = (-3, 4)$ and

$$f(-3, 4) = -25.$$

Therefore

$$\boxed{f_{\max} = 25 \text{ at } (3, -4)}, \quad \boxed{f_{\min} = -25 \text{ at } (-3, 4)}.$$

Question 4

The curves $y = x^2$ and $y = 2x$ intersect when

$$x^2 = 2x \implies x = 0, 2.$$

For $0 \leq x \leq 2$,

$$x^2 \leq y \leq 2x.$$

Thus

$$V = \int_0^2 \int_{x^2}^{2x} (xy + 4) dy dx.$$

Then

$$V = \int_0^2 \left[\frac{x}{2} y^2 + 4y \right]_{x^2}^{2x} dx = \int_0^2 \left(2x^3 + 8x - \frac{1}{2} x^5 - 4x^2 \right) dx.$$

Therefore

$$V = \left[\frac{1}{2} x^4 + 4x^2 - \frac{1}{12} x^6 - \frac{4}{3} x^3 \right]_0^2 = \boxed{8}.$$

Practice Test 3

Question 1

a) The denominator is nonzero at $(-1, 2)$, so

$$\lim_{(x,y) \rightarrow (-1,2)} \frac{x^2 + xy + y^2}{1 + x^2 + y^2} = \frac{1 - 2 + 4}{1 + 1 + 4} = \boxed{\frac{1}{2}}.$$

b) Along $y = 0$,

$$\frac{x^2 - y^2}{x^2 + y^2} = 1.$$

Along $x = 0$,

$$\frac{x^2 - y^2}{x^2 + y^2} = -1.$$

The limit does not exist.

c) Using polar coordinates,

$$\frac{x^3 y}{x^2 + y^2} = r^2 \cos^3 \theta \sin \theta \rightarrow 0.$$

Thus

$$\boxed{a = 0}.$$

Question 2

a)

$$z = e^{xy} + x \sin y, \quad x = r + t^2, \quad y = rt.$$

At $(r, t) = (0, 1)$, $(x, y) = (1, 0)$. Also

$$z_x = ye^{xy} + \sin y, \quad z_y = xe^{xy} + x \cos y.$$

At $(1, 0)$,

$$z_x = 0, \quad z_y = 2.$$

Also

$$x_r = 1, \quad y_r = t = 1, \quad x_t = 2t = 2, \quad y_t = r = 0.$$

Thus

$$\boxed{\frac{\partial z}{\partial r} = z_x x_r + z_y y_r = 2}, \quad \boxed{\frac{\partial z}{\partial t} = z_x x_t + z_y y_t = 0}.$$

b) Let

$$F(x, y, z) = x \cos z + y \sin z - 2 = 0.$$

Then

$$F_x = \cos z, \quad F_y = \sin z, \quad F_z = -x \sin z + y \cos z.$$

Therefore

$$\boxed{\frac{\partial z}{\partial x} = \frac{\cos z}{x \sin z - y \cos z}}, \quad \boxed{\frac{\partial z}{\partial y} = \frac{\sin z}{x \sin z - y \cos z}}.$$

c)

$$f(x, y) = \sqrt{x^2 + 2y^2}, \quad \nabla f = \left\langle \frac{x}{\sqrt{x^2 + 2y^2}}, \frac{2y}{\sqrt{x^2 + 2y^2}} \right\rangle.$$

At $(2, 1)$,

$$\nabla f(2, 1) = \left\langle \frac{2}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right\rangle.$$

The direction from $(2, 1)$ to $(-1, 5)$ is $\langle -3, 4 \rangle$, so

$$\mathbf{u} = \left\langle -\frac{3}{5}, \frac{4}{5} \right\rangle.$$

Hence

$$D_{\mathbf{u}}f(2, 1) = \left\langle \frac{2}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right\rangle \cdot \left\langle -\frac{3}{5}, \frac{4}{5} \right\rangle = \boxed{\frac{2}{5\sqrt{6}}}.$$

Question 3

$$f(x, y) = x^2 + 2y^2 - 4x.$$

Interior critical points satisfy

$$f_x = 2x - 4 = 0, \quad f_y = 4y = 0,$$

so $(x, y) = (2, 0)$, which lies inside $x^2 + y^2 \leq 9$. Then

$$f(2, 0) = -4.$$

On the boundary $x^2 + y^2 = 9$, use Lagrange multipliers:

$$\langle 2x - 4, 4y \rangle = \lambda \langle 2x, 2y \rangle.$$

Thus

$$x - 2 = \lambda x, \quad 2y = \lambda y.$$

If $y = 0$, then $x = \pm 3$, giving

$$f(3, 0) = -3, \quad f(-3, 0) = 21.$$

If $y \neq 0$, then $\lambda = 2$, so

$$x - 2 = 2x \implies x = -2.$$

Using $x^2 + y^2 = 9$ gives $y = \pm\sqrt{5}$, and

$$f(-2, \pm\sqrt{5}) = 4 + 10 + 8 = 22.$$

Therefore

$$\boxed{f_{\min} = -4 \text{ at } (2, 0)}, \quad \boxed{f_{\max} = 22 \text{ at } (-2, \pm\sqrt{5})}.$$

Question 4

The curves $y = x^2$ and $y = 2 - x^2$ intersect when

$$x^2 = 2 - x^2 \implies x = \pm 1.$$

For $-1 \leq x \leq 1$,

$$x^2 \leq y \leq 2 - x^2.$$

Thus

$$V = \int_{-1}^1 \int_{x^2}^{2-x^2} (1 + x^2 + y) \, dy \, dx.$$

The inner integral gives

$$V = \int_{-1}^1 (4 - 2x^2 - 2x^4) \, dx.$$

Therefore

$$V = 2 \int_0^1 (4 - 2x^2 - 2x^4) \, dx = 2 \left(4 - \frac{2}{3} - \frac{2}{5} \right) = \boxed{\frac{88}{15}}.$$

Practice Test 4

Question 1

a)

$$\frac{e^{xy} - 1}{x} = y \cdot \frac{e^{xy} - 1}{xy}.$$

As $(x, y) \rightarrow (0, 2)$, $xy \rightarrow 0$, so

$$\frac{e^{xy} - 1}{xy} \rightarrow 1.$$

Therefore

$$\lim_{(x,y) \rightarrow (0,2)} \frac{e^{xy} - 1}{x} = 2(1) = \boxed{2}.$$

b) Along $y = mx^2$,

$$\frac{x^2 y}{x^4 + y^2} = \frac{mx^4}{x^4 + m^2 x^4} = \frac{m}{1 + m^2}.$$

Since this depends on m , the limit does not exist.

c) Let $r^2 = x^2 + y^2$. Then

$$\frac{1 - \cos(x^2 + y^2)}{x^2 + y^2} = \frac{1 - \cos(r^2)}{r^2}.$$

Since $1 - \cos u \sim u^2/2$ as $u \rightarrow 0$,

$$\frac{1 - \cos(r^2)}{r^2} \rightarrow 0.$$

Thus

$$\boxed{a = 0}.$$

Question 2

a)

$$w = x \ln y + y \cos x, \quad x = pq, \quad y = p + q.$$

At $(p, q) = (1, 1)$, $(x, y) = (1, 2)$. Also

$$w_x = \ln y - y \sin x, \quad w_y = \frac{x}{y} + \cos x.$$

So

$$w_x = \ln 2 - 2 \sin 1, \quad w_y = \frac{1}{2} + \cos 1.$$

Also

$$x_p = q = 1, \quad y_p = 1, \quad x_q = p = 1, \quad y_q = 1.$$

Therefore

$$\boxed{\frac{\partial w}{\partial p} = \ln 2 - 2 \sin 1 + \frac{1}{2} + \cos 1},$$

and

$$\boxed{\frac{\partial w}{\partial q} = \ln 2 - 2 \sin 1 + \frac{1}{2} + \cos 1}.$$

b) Let

$$F(x, y, z) = x^2z + yz^2 - 6 = 0.$$

Then

$$F_x = 2xz, \quad F_y = z^2, \quad F_z = x^2 + 2yz.$$

Thus

$$\boxed{\frac{\partial z}{\partial x} = -\frac{2xz}{x^2 + 2yz}}, \quad \boxed{\frac{\partial z}{\partial y} = -\frac{z^2}{x^2 + 2yz}}.$$

c)

$$f(x, y) = \sin(xy), \quad \nabla f = \langle y \cos(xy), x \cos(xy) \rangle.$$

At $(\pi/2, 1)$,

$$\cos(xy) = \cos(\pi/2) = 0.$$

Therefore

$$\nabla f(\pi/2, 1) = \langle 0, 0 \rangle,$$

so every directional derivative is

$$\boxed{0}.$$

Question 3

The only interior critical point of $f(x, y) = xy$ is $(0, 0)$, where $f = 0$. On the boundary $x^2 + y^2 = 8$, use Lagrange multipliers:

$$\nabla f = \lambda \nabla g, \quad \langle y, x \rangle = \lambda \langle 2x, 2y \rangle.$$

Thus

$$y = 2\lambda x, \quad x = 2\lambda y.$$

For nonzero x, y , $\lambda = \pm 1/2$. If $\lambda = 1/2$, then $y = x$, and

$$2x^2 = 8 \implies x = \pm 2.$$

This gives $(2, 2)$ and $(-2, -2)$, where $xy = 4$. If $\lambda = -1/2$, then $y = -x$, giving $(2, -2)$ and $(-2, 2)$, where $xy = -4$. Therefore

$$\boxed{f_{\max} = 4 \text{ at } (2, 2), (-2, -2)}, \quad \boxed{f_{\min} = -4 \text{ at } (2, -2), (-2, 2)}.$$

Question 4

The region is enclosed by

$$x = y^2 - 1, \quad x = 3 - y^2.$$

The intersections satisfy

$$y^2 - 1 = 3 - y^2 \implies 2y^2 = 4 \implies y = \pm\sqrt{2}.$$

For $-\sqrt{2} \leq y \leq \sqrt{2}$,

$$y^2 - 1 \leq x \leq 3 - y^2.$$

Thus

$$V = \int_{-\sqrt{2}}^{\sqrt{2}} \int_{y^2-1}^{3-y^2} (3-x+y^2) \, dx \, dy.$$

The inner integral simplifies to

$$\int_{y^2-1}^{3-y^2} (3-x+y^2) \, dx = 8-2y^4.$$

Therefore

$$V = \int_{-\sqrt{2}}^{\sqrt{2}} (8-2y^4) \, dy = 16\sqrt{2} - \frac{16\sqrt{2}}{5} = \boxed{\frac{64\sqrt{2}}{5}}.$$